

Chapter 3. Financial Identification Systems

A key aspect of financial data, such as prices and transactions, is that it can provide informational value only if we can reliably assign each record to its corresponding entity. Being able to filter a dataset to get data for a specific entity unlocks the ability to analyze the data in meaningful ways.

To this end, financial market participants have developed and employed different types of financial identifiers. Nevertheless, data identification remains notably challenging and is widely regarded as one of the most critical problems in financial data management. The outstanding issue of reference data management, presented in the previous chapter, fundamentally revolves around financial identification and the matching of various identifiers that reference the same financial market entity.

This chapter will discuss the problem of financial data identification, illustrate the desired properties of financial identification systems, and examine the key features and limitations of current systems.

If you are going to become a financial data engineer, dealing with financial identifiers and knowing how to manage their shortcomings will be one of the main challenges you will regularly face. So, let's dive into this issue.

Financial Identifiers

The predominantly digital nature of financial market operations and transactions necessitates recording and querying them through information and database systems. At its heart, a reliable financial information

system is an identification system: a way of telling who interacts with whom, a way of distinguishing one financial entity from another, and a way of finding all records that belong to the same entity. Consequently, well-identified financial data will deliver valuable insights and a significant competitive edge. To learn more, let's further detail what financial identifiers and identification systems are, why they are essential for financial markets, and who creates and maintains financial data identification systems.

Financial Identifier and Identification System Defined

From our discussion so far, you've seen how financial data is produced in large volumes and various formats, which are subsequently organized into more coherent collections known as financial datasets. Typically, each financial dataset must have an *observation unit* (or *statistical unit*) that represents the object for which information (*data points*) is available. For example, a company fundamentals dataset would have the company as the unit of observation. To distinguish data points for one unit from the others, such as company A's data from company B's, a data identifier (company identifier in our example) should be attached to each data point. Without an identifier, financial datasets would be of no practical use. [Figure 3-1](#) illustrates the concept. The tables shown in the figure convey information about the annual revenues of different companies observed at multiple periods. The data in the left table is not very useful as it is hard to determine which firm the statistics are referencing. When the `company_id` identifier is added in the right table, it becomes possible to distinguish between several companies that the data references.

Unidentified data		Identified data		
Date	Annual revenues	company_id	Date	Annual revenues
31-12-2020	\$2,100,000	company_a	31-12-2020	\$2,100,000
31-12-2021	\$1,880,790	company_a	31-12-2021	\$1,880,790
31-12-2022	\$2,250,000	company_a	31-12-2022	\$2,250,000
31-12-2023	\$2,405,000	company_a	31-12-2023	\$2,405,000
31-12-2022	\$1,600,00	company_b	31-12-2022	\$1,600,00
31-12-2020	\$500,000	company_b	31-12-2020	\$500,000
31-12-2019	\$20,000,000	company_c	31-12-2019	\$20,000,000

Figure 3-1. Unidentified dataset (left) versus identified dataset (right)

In this book, I define a financial identifier and financial identification system as follows:

A financial identifier is a character sequence associated with a particular financial entity (e.g., company, individual, transaction, asset, document, sector group, event, etc.), enabling accurate identification of said entity across one or more financial datasets or information systems. A financial identifier can be any combination of numeric digits (0-9), alphabet letters (a-z, A-Z), and symbols. A financial identification system creates principles and procedures for generating, interpreting, storing, assigning, and maintaining financial identifiers.

There are a few more key terms that are important to establish. First, calling an identifier a code, ID, or symbol is common. Second, financial identification systems can generate identifiers following an *encoding* system or as *arbitrary IDs*. An encoding system converts words, letters, numbers, and symbols into a short, standardized format for identification, communication, and storage. The reverse process is decoding, which converts the code sequence back to its original form to make it easier to understand. Identifiers that do not adhere to an encoding system are often described as arbitrary; they are randomly created and assigned and have no particular meaning. Third, the field that deals with building financial identification systems is frequently referred to as *symbolology*, a term you will often hear when learning and working with financial identifiers.

The Need for Financial Identifiers

Financial data identifiers serve various purposes, one of the most common being the identification of financial instruments and entities involved in market transactions. This type of identification data is often referred to as reference data, as we discussed in [Chapter 2](#).

In any financial transaction, it is crucial to include the identifiers of the exchanged instruments and the entities participating in the agreement. Additionally, the transaction itself may be assigned an identifier for tracking purposes. As a result, identifiers are integral throughout the entire lifecycle of a financial transaction, from pre-trade to trade to post-trade settlement. They facilitate swift and efficient market transactions, enhance communication among market participants, increase transparency, and reduce operational costs and errors.¹

Another important use of financial identifiers is for reporting purposes. Regulatory authorities can demand various information from financial institutions, such as market exposure, capital, risk concentration, liquidity, assets and liabilities, and trades. Financial institutions need to aggregate data from multiple sources and divisions to prepare the required report. Financial identifiers are crucial in this situation since they enable data collection and consolidation, ensuring accurate and timely reporting. In addition, by adding identifiers to the reported data, regulators would find it simple to examine the information and judge the reporting institution's compliance. Several regulatory frameworks, such as MiFID II, mandate that reporting institutions use certain financial identities when reporting data. This, in turn, imposes an extra obligation on reporting institutions to establish a reliable financial identification system.²

Financial identifiers are essential for exchange listing and trading purposes. To list and trade a financial security on a trading venue, it must be assigned an identifier. This enables investors, traders, and market makers to easily locate, track, buy, sell, and analyze financial instruments.

Last but not least, financial identifiers are required as an essential data field when performing financial data analysis. A substantial portion of fi-

nancial analysis is cross-sectional, where the focus is on studying the behavior and differences among different financial entities (e.g., assets, companies, etc.). Using financial identities would enable the analyst to pick the right data sample, run quality checks and filters, eliminate duplicates, and match the same entity across numerous datasets.

Who Creates Financial Identification Systems?

Various organizations, spanning both public and private sectors, may generate and assign a financial identifier. Some organizations develop recommendations for financial identification systems but do not issue the identifier for those who require it; others issue identifiers based on existing standards or recommendations; and still others develop and issue the identifier. Let's explore this variety of roles and functions with some examples.

International Organization for Standardization (ISO)

The *International Organization for Standardization* (ISO) is an independent organization that creates and promotes voluntary and consensus-based international standards for various technical and nontechnical fields. It is composed of representatives from the national standards organizations of 169 countries.

Throughout the years, the ISO has demonstrated considerable interest and involvement in developing international financial identifiers. For example, the ISO standard known as the International Securities Identification Number (ISIN) has emerged as the leading identifier in international security trading, clearing, and settlement. Later in this chapter, I will cover the ISIN identifier as well as other ISO-based identifiers in detail.

Crucially, the ISO does not issue and assign identifiers for market participants; instead, this job is delegated to so-called National Numbering Agencies, which we will examine in the following section.

Familiarity with industry standards is essential for a financial data engineer to excel. However, how do organizations such as the ISO create a standard? As reported by the official [website](#), the ISO does not unilaterally decide when to create a standard; instead, it responds to a market need raised by stakeholders such as companies, consumer associations, academia, NGOs, and government and consumer groups. A typical scenario involves an industry group reporting the need for a standard to its national member, who subsequently approaches the ISO.

Once a market need for a standard emerges, the ISO appoints a committee composed of independent technical experts nominated by ISO members. A committee may have subcommittees and working groups. For example, ISO/TC 68 is the ISO committee tasked with overseeing financial services standards globally, and it has three main subcommittees:

ISO/TC 68/SC 2

Covers information security in financial services

ISO/TC 68/SC 8

Covers reference data for financial services

ISO/TC 68/SC 9

Covers information exchange for financial services

The nominated committee starts the process by discussing the nature, scope, and key elements of the standard and submitting a draft proposal that meets the market need. The draft is then shared for further review and recommendations. Reaching a final agreement is consensus based and relies on a voting mechanism. If a consensus is not reached, then the draft will be modified and voted on again. Developing an ISO standard typically takes three years from initial proposal to final publication.

National Numbering Agencies

A *National Numbering Agency* (NNA) is a national organization that issues and promotes ISO-based financial identifiers. Each country is free to assign the role of NNA to a local market player, which can be a stock exchange, central bank, regulator, clearing house, financial data provider, or custodian. For example, the UK NNA is the London Stock Exchange, and Luxembourg's is Clearstream Banking (a central securities depository). Some countries don't have an NNA; in such cases, a Substitute Numbering Agency (SNA) is appointed. Examples of SNAs include CUSIP Global Services in the US and WM Datenservice in Germany.

The Association of National Numbering Agencies (ANNA) was established to coordinate between the different national NNAs. ANNA collects and aggregates identifier data from all its members into a global centralized dataset called the ANNA Service Bureau (ASB) to ensure data quality and guarantee global interoperability.

Financial data vendors

Most financial data vendors create their own identification systems to support the development of their products and services. For instance, when aggregating data from many sources, the data vendor might face the issue of identifier heterogeneity, which can be hard to resolve. It may also be the case that the data has no identifier, for example, when working with unstructured data such as news and text. Additionally, external identifiers might lack some of the properties the vendor needs (e.g., uniqueness). In such cases, creating a new vendor-based identification system is a common practice. Vendors such as S&P Global Market Intelligence, LSEG, Bloomberg, and FactSet all have their own in-house identifiers.

In certain instances, financial data vendors develop their identifiers because they create, rather than aggregate, their content. For example, news analytics providers such as RavenPack create and provide structured news and event data from unstructured content for a large number of entities. Since RavenPack independently extracts entities from text, it

has developed its proprietary unique entity identifier system, known as the RavenPack Unique Entity Identifier.

Finally, given the increasing value of financial identifier data (e.g., for reference data management), financial data vendors may find it a profitable opportunity to create and license their financial identifiers.

Financial institutions

Financial institutions create identifiers to facilitate internal processes, such as transactions, account management, client identification, and payment card number generation. Additionally, they might collect financial data from various sources, both internal and external, which may come with different identifiers. In these cases, a financial institution may opt to create its internal identification system to allow for a flexible design that matches business requirements and enables easy data retrieval and aggregation.

In conclusion, financial identifiers are critical to the efficiency, interoperability, and transparency of financial markets. All market participants deal with identifiers, either as users or as participants in their development. At this point, you may be wondering why it is so difficult to design a financial identification system that works in all circumstances and meets all market demands. To answer this question, we must first examine the desirable attributes of financial identification systems and understand the various constraints and challenges involved. This is what I'll talk about next.

Desired Properties of a Financial Identifier

Designing and maintaining a good financial identification system is one of the most persistent challenges in the financial industry. The difficulty stems from the need to balance various desirable properties, each of which may conflict with others. In this section, I build upon the framework proposed in the seminal work of Jules J. Berman, *Principles and*

Practice of Big Data (Academic Press), to derive a minimal set of desired properties for a financial identification system. These properties include uniqueness, globality, scalability, completeness, accessibility, authenticity, granularity, permanence, immutability, and security. Let's explore each property in detail.

Uniqueness

Uniqueness refers to the quality of being one of a kind. For instance, fingerprints are unique to each human; no two humans on earth can have the same fingerprint. In the same way, a financial identification system must uniquely identify financial entities and never assign the same identifier to distinct entities. [Figure 3-2](#) illustrates the concept of uniqueness. The identification system on the left is not unique since it assigns the identifier 56HF128 to two separate entities: WTK Inc. and XYZ Inc. However, the system on the right is unique as it assigns a distinct identifier to all three financial entities.

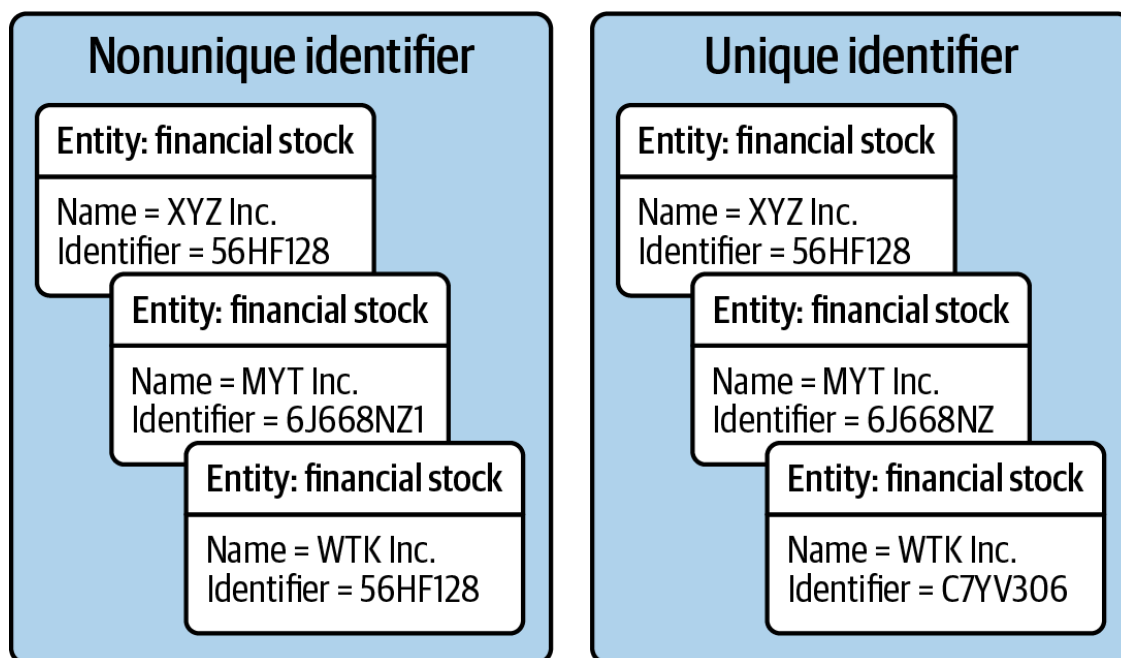


Figure 3-2. Unique (right) versus nonunique (left) identification systems

Crucially, the concept of a unique identifier can become ambiguous when applied to financial markets. To illustrate how, let's consider the following scenarios:

- A company is listed on two stock exchanges, A and B. Should the listed stock have two identifiers, one for each market, or a single identifier for both?
- A financial instrument is trading in two countries; do we need separate identifiers for each country?
- A financial instrument can have multiple issues. Should we treat each issue as a unique instrument or use a single identifier for all?
- Following a company merger or acquisition, does the resulting entity represent a new, unique company or maintain the identity of the original?
- A financial transaction contains multiple instructions that relate to various instruments. Should each instruction have its own identifier or simply use the parent transaction identifier?

Quite challenging, no? Now, I imagine that you might have a solution for the above scenarios. But this is exactly where the issue lies; different market participants make different assumptions about what a unique financial identifier should be, and this has led to the development and adoption of numerous identifiers that differ in the way they define uniqueness. This can explain the common practice of using multiple identifiers within a dataset, for example, to identify the security with one identifier and the market where it is traded with another. If you are designing a financial identification system, my advice for you is to carefully think about the concept of uniqueness and discuss it with your business team to avoid serious shortcomings that might arise later.

Globality

Financial markets are complex and dynamic systems that are in continuous evolution and expansion. To illustrate how, let's consider the following facts:

- New companies are established and listed on the market.
- New financial instruments are created, listed, and traded in different markets and jurisdictions.
- Trading activities take place in various venues, such as centralized exchanges, trading platforms, and OTC markets.

- Consumer demand and preferences for financial products and services are constantly evolving, creating opportunities for the introduction of new offerings.
- Financial transactions span multiple countries, markets, and jurisdictions.
- New markets are established for exchanging new financial products.
- New financial entities are recognized and extracted from financial data.
- New exchange platforms and mechanisms gain popularity (e.g., cryptocurrencies).
- New types of data on financial activities are recorded and consumed.
- New financial regulatory requirements are published and enforced.
- New market standards are released, promoted, and adopted.
- New financial technologies emerge and diffuse among market participants.

These dynamics, among others, have contributed to a significant increase in the number and variety of financial entities that require identification. To meet this demand, a financial identification system must be able to accommodate the expanding ecosystem of financial activities and entities. I call such property *globality*. For example, a global identification system can do the following things:

- Expand from assigning identifiers at a national level to an international scale
- Expand its scope from assigning identifiers solely within centralized markets to OTC and other types of markets
- Expand its coverage to include various financial instruments such as indices, derivatives, digital assets, and loans

Figure 3-3 illustrates a straightforward example showcasing the concept of globality. The system on the top is global, as it expands its coverage to new areas that emerge from market expansion. The system on the bottom is nonglobal, as it covers only three areas and can't expand further.

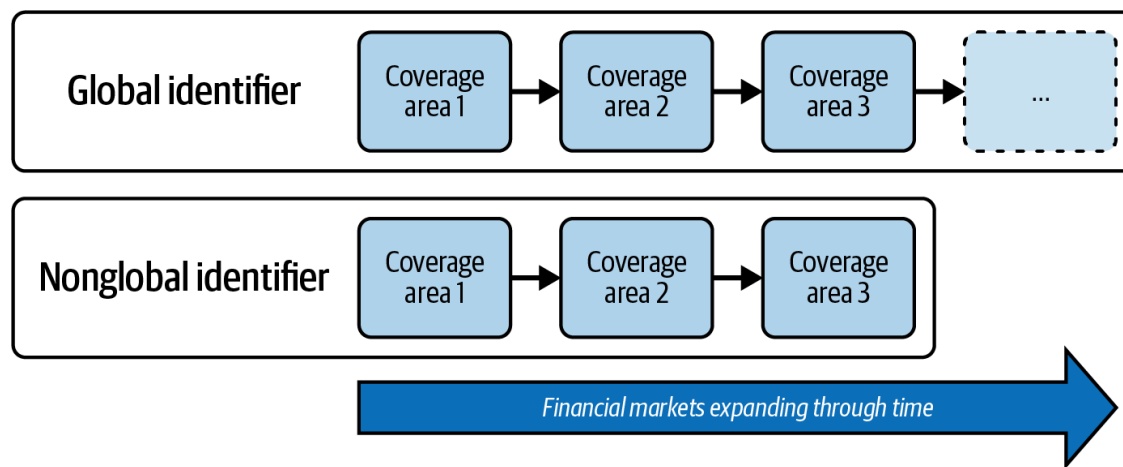


Figure 3-3. Global (top) versus nonglobal (bottom) identification systems

Several identifiers, in particular vendor-specific identifiers, are limited to certain markets (e.g., stocks, bonds), exchanges (e.g., NYSE trades and quotes), or jurisdictions (e.g., US stocks, UK stocks). However, some identifiers have been developed to cover a broader range of market entities. The best example is perhaps Bloomberg’s Financial Instrument Global Identifier (FIGI), which we will discuss in more detail later in this chapter.

Scalability

A scalable financial identification system does not exhaust its pool of assignable identifiers. Several reasons could lead to a shortage in the available supply of identifiers in a financial identification system:

Rapid market growth

If the number of financial entities requiring identifiers increases rapidly, the identification system may struggle to keep up with the demand. One example is the issuance of a high volume of short-term financial instruments, such as commercial papers, repurchase agreements, and certificates of deposit. These instruments typically have short maturities and are frequently rolled over, creating a constant demand for new identifiers. If each instrument requires a unique identifier, it can quickly reduce the available pool of new identifiers.

Limited character length

If the identifier format is limited in character length, exclusively employs numeric characters, or imposes format constraints, there may be a finite number of possible combinations, causing the identification system to exhaust its identifiers soon. For example, the Issuer Identification Number (IIN), presented later in this chapter, was expanded from a six-digit to an eight-digit format due to a [shortage in the supply of assignable identifiers](#). Another example is the SEDOL identifier, discussed later in this chapter, which was changed from a numeric to an alphanumeric format following plans to expand its market coverage.

Poor allocation strategy

If the identification system does not allocate the identifiers optimally, it might lead to early exhaustion. Examples of poor allocation strategies include the following:

Category-based allocation

Allocating identifiers based on specific categories (e.g., different types or classes of stocks, bonds, derivatives) can lead to the exhaustion of available identifiers for a particular category if it grows more quickly than others. For instance, if bond instruments are allocated identifiers from 00000-29999 and stock instruments from 30000-49999, the bond category might run out of identifiers much earlier if there is a surge in bond issuance.

Reserved ranges

Reserving large ranges of identifiers for future use or specific purposes, such as special market events, regulatory reporting, or new financial instruments, can significantly reduce the pool of identifiers available for general use.

Addressing these issues entails proactive planning, market growth projection, periodic evaluations, and adaptability of the identification system to the ever-changing financial market ecosystem.

Completeness

Completeness requires that an identifier be assigned to each uniquely identifiable entity covered by the identification system. In other words, if an identification system is created to identify a set of financial entities, then each entity in the set must have an identifier. [Table 3-1](#) illustrates the concept by comparing complete and incomplete identifiers. Identification system A is incomplete because it lacks identifiers for entities 3, 4, and 6. In contrast, Identification system B is complete, as it assigns identifiers to all six entities.

Table 3-1. Comparing complete and incomplete identifiers

Entity	Identifier A (incomplete)	Identifier B (complete)
Entity 1	19982243	A5J234HS
Entity 2	87987924	B5J874GS
Entity 3	NULL	T3H7Z589
Entity 4	NULL	GQ16B437
Entity 5	23987912	N9M3F16S
Entity 6	NULL	K485GV1Z

A financial identification system may suffer from incompleteness for various reasons. For instance, a newly established company identification system might not backfill the history of its covered universe because some companies have failed, gone bankrupt, delisted, merged with other companies, or changed their names. Another common scenario arises when merging different datasets with different identifiers. For example, if you join data using a US-based identifier with data using a global identifier, many entities might lack the US identifier.

Accessibility

Financial identifier data should be accessible, meaning it should not be restricted by license fees or usage limits, or monopolized by market players. Limited access to financial identifiers can result in market inefficien-

cies and a lack of transparency when conducting transactions and reporting.³ Logically, there are situations where financial identifiers, like credit card numbers or bank account IDs, must be secured.

In some cases, financial identifiers are freely available (e.g., on online trading platforms or stock exchange websites). However, collecting and aggregating this data in a structured format for a large number of entities can be overwhelming. Financial data vendors provide such products, often sold as reference data, but these are often subscription based.

Some market players launched initiatives to promote open-access financial identifiers. For example, LSEG openly [released its Permanent Identifier \(PermID\) system](#), which provides comprehensive coverage across a wide variety of entity types, including companies, financial instruments, issuers, funds, and people. Another example is OpenFIGI, an open-access system that allows anyone to request Bloomberg's Financial Instrument Global Identifier (FIGI), which I discuss later in this chapter.

Timeliness

The timeliness of a financial identification system refers to its ability to process and generate identifiers quickly and efficiently. When a new financial entity enters the market or is created within a system (e.g., a new issue of a financial instrument), a timely financial identification system should do the following:

- Enable market participants to request an identifier quickly, ideally in real or near-real time
- Process requests quickly, allowing the issuing entity to allocate identifiers promptly
- Make accessible the newly allocated identifiers to other market participants without delay

Efficient and timely generation and dissemination of financial identifiers are essential for enhancing the efficiency of financial market operations and transactions.

Authenticity

An authentic financial identification system can confirm whether a specific identifier was generated by it. To draw an analogy, this is the same as telling whether a watch displaying the label *Rolex* is indeed an authentic Rolex.

Financial identifiers are often designed following a specific symbology specification, which can be a formula or standardized format. This, in turn, enables the development of programs that can verify the identifier's adherence to these specifications. A common practice involves the addition of a *check digit* to the identifier string. When a check digit is included, typically at the end of the identifier, an algorithm can be used to authenticate the identifier based on this digit. Later in this chapter, I will illustrate various examples of identifier check digits and their calculation logic.

Granularity

In financial markets, it is quite common to observe hierarchical arrangements and structures within and between entities:

- Asset origination chain (e.g., issuing company → securities → stocks → common stocks → issues).
- Trading context (e.g., international → continental → national → market).
- Pyramid organizational structures (e.g., CEO → president → vice-president → middle management → team leaders → employees) .
- Company ownership structures (e.g., parent company A owns B (child), B owns D, and D owns F, G, K).
- Asset classifications, where assets are organized in groups, categories, and subcategories (e.g., derivatives → options → stock options → call stock option).
- Sector classifications, which structure economic sectors using an industry taxonomy (e.g., sectors → industry groups → industries → sub-industries).
- Packaged transactions such as syndicated loans (package → facilities).

- Complex financial instruments such as multi-leg options or interest rate swaps, where the same instrument has multiple child instruments.
- In the fund industry, an umbrella fund structure allows for the creation of multiple subfunds, each with distinct investment strategies, further divided into various share classes tailored to different investor needs.⁴

A financial identification system is *granular* if it can scale to provide different levels of details that reflect market hierarchies. [Figure 3-4](#) illustrates this concept. In this example, the entity requiring identification should be recognizable at international, national, and market levels. The identification system on the left is nongranular because it uses the same identifier for the entity across all levels. In contrast, the system on the right is granular because it assigns a unique identifier to the entity at each level.

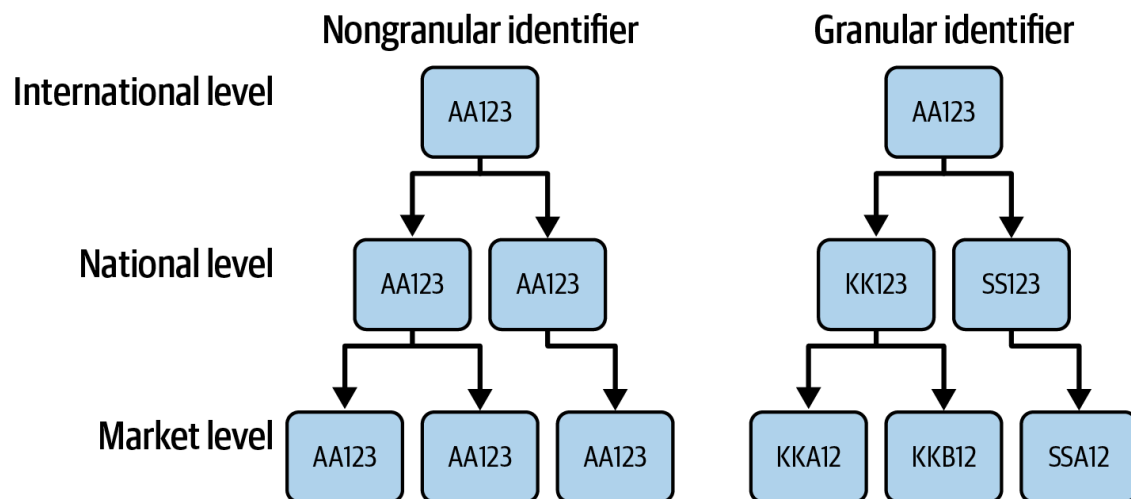


Figure 3-4. Granular versus nongranular financial identification systems

Permanence

A reliable financial identification system must ensure that identifiers and their associated data are permanent and durable. This is essential for guaranteeing market trust and confidence and allowing the tracking and referencing of financial instruments and transactions over time. For example, a bank client should be able to go to the bank at any time and ask for a statement about their past transactions. If the customer closes their

accounts with the bank, their identification data should not disappear from the system.

In some cases, a financial identifier can be interrupted; for example, a commercial data vendor might stop maintaining an identifier as they develop a new identification system or switch to using an industry-wide system. Additionally, an identification system might be replaced with a new one if it fails to meet certain desired properties, such as uniqueness, globality, and granularity. The concept of permanent versus nonpermanent identifiers is shown in [Figure 3-5](#). The identification system on the left is permanent as it could return the associated data at time t and time $t+n$. In contrast, the system on the right is nonpermanent as it returns data at time t but returns none at time $t+n$.

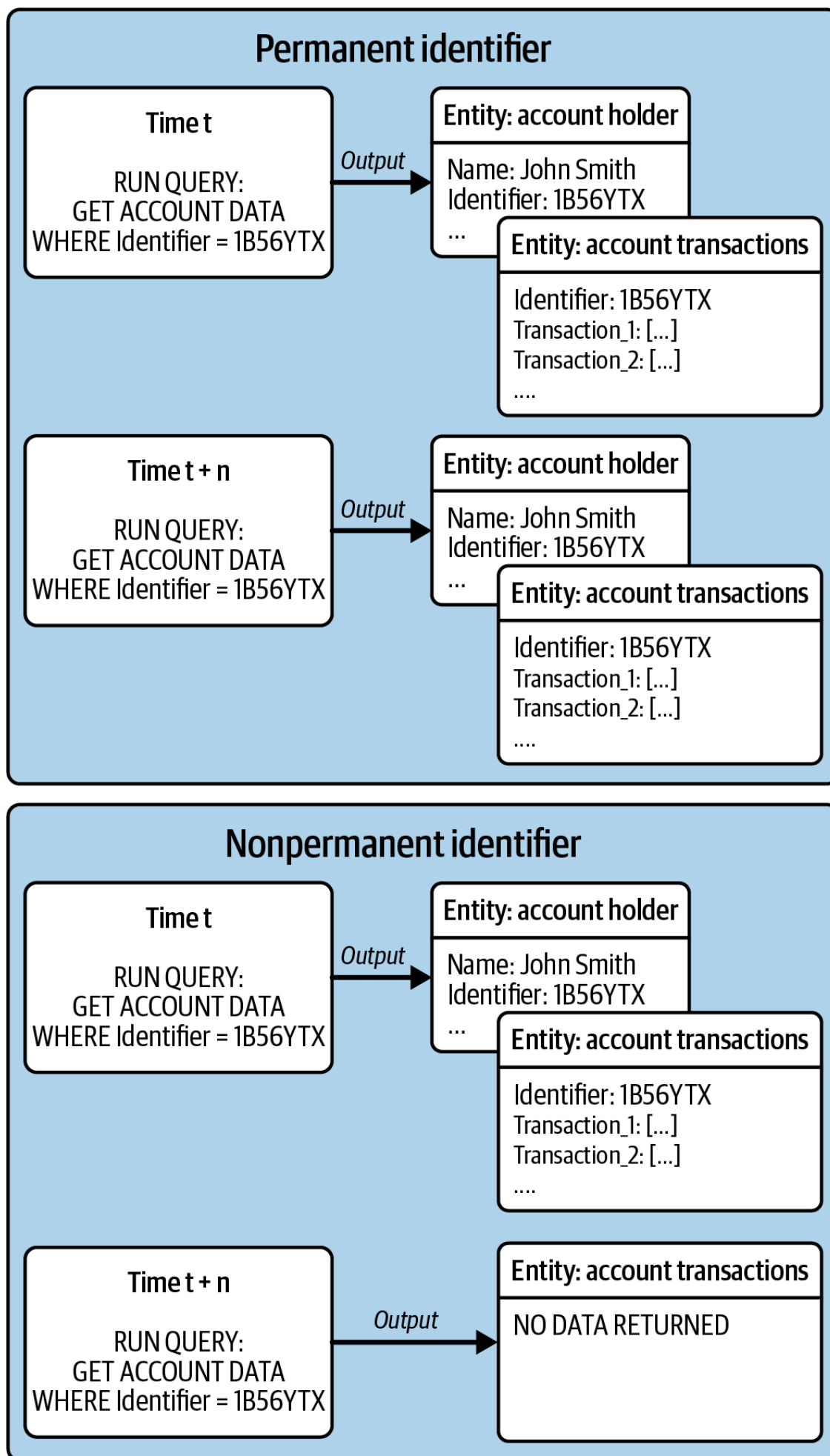


Figure 3-5. Permanent versus nonpermanent identification systems

Immutability

A robust financial identification system must ensure the immutability of its identifiers over time. This means that an identifier associated with a specific entity should neither change nor be reassigned to another entity. If the identified entity ceases to exist, its identifier should never be reused. Such a property is essential for maintaining transparency, traceability, and effective monitoring.

Besides temporal immutability, identifiers must persist through all stages of financial activities. For example, a transaction’s lifecycle, which typically comprises multiple steps from initiation, through execution, to settlement, should consistently use the same identifier. [Table 3-2](#) illustrates the concept of mutable versus immutable identifiers. Identifier A is immutable, as all three entities have the same identifier in 2005 and 2022. Identifier B is not mutable, as entities 1 and 3 changed their identifier in 2022. Moreover, the identifier of entity 1 used in 2005 was reassigned to entity 3 in 2022.

Table 3-2. Mutable versus immutable identifiers

Entity	Year	Identifier A (immutable)	Identifier B (mutable)
Entity 1	2005	XZY743	XZY743
Entity 2	2005	ABC376	ABC376
Entity 3	2005	MNT098	MNT098
Entity 1	2022	XZY743	KKH654
Entity 2	2022	ABC376	ABC376
Entity 3	2022	MNT098	XZY743

NOTE

Corporate events such as mergers, acquisitions, spin-offs, corporate restructurings, and rebranding might change a company’s structure, potentially resulting in the creation of a new entity. In these cases, a new identifier may be assigned.

Security

Security must be prioritized when designing a financial identification system, as these systems are vulnerable to malicious attacks. Consider the damage that an attacker could inflict, for instance, if they succeeded in hacking an identification system and irreversibly changing its identifiers. Even worse is the scenario in which sensitive identifying data is compromised and utilized to commit financial crimes against the entities identified in the data.

To summarize, developing an optimal financial identification system is challenging since it involves multiple complex properties that may be difficult to achieve in a single system. For this reason, financial markets have developed a plethora of financial identification systems, each serving a particular purpose and offering unique characteristics. Some of these systems are widely adopted, while others are gaining attention in response to evolving market needs. The next section will provide you with an overview of current financial identification systems.

Financial Identification Systems Landscape

The current landscape of financial identification systems is extensive, comprising open source and proprietary systems designed with different properties and for various purposes. Several standardization efforts have been initiated, yet a universally accepted standard has not been established. To give you an idea of the current scale, a Wikipedia [search for “security identifier”](#) yields results for 16 different identifiers. The ISO conducted an [inventory of the current national and international financial identification standards](#) and identified a total of 19 different systems. These identifiers and several others are illustrated in [Figure 3-6](#).

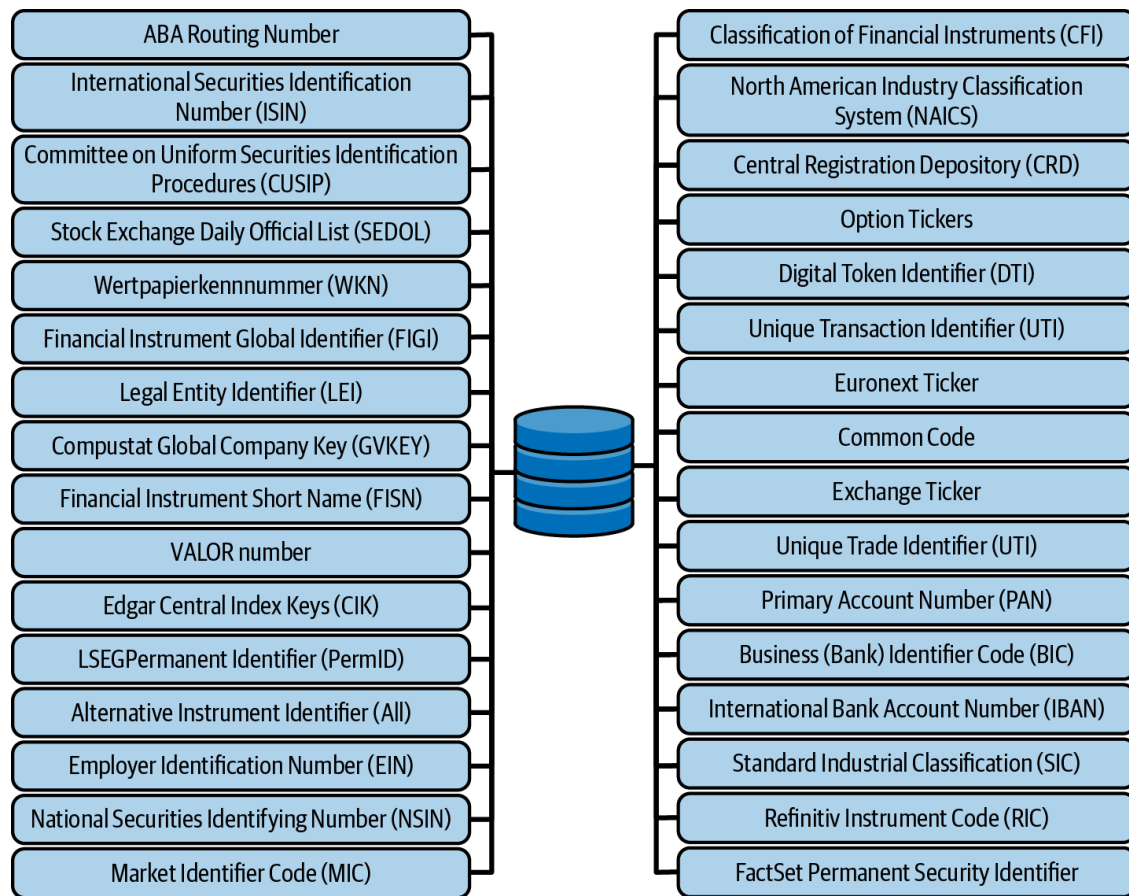


Figure 3-6. Financial identification systems in use globally

In the following sections, I will discuss some of the most important financial identification systems used by markets, along with their key characteristics and challenges.

International Securities Identification Number

The *International Securities Identification Number* (ISIN) is a 12-character alphanumeric code, defined in [ISO 6166](#), that uniquely identifies a wide range of financial securities such as stocks, bonds, derivatives, fund share classes, and indexes.

In principle, an ISIN should represent a set of fungible financial instruments. For instance, the common stocks of a given company are fungible with each other, regardless of the issuance date, as they all share the same specifications. In this scenario, the same ISIN is typically used to represent the company's common stock. However, if the company issues different classes of stocks, such as preferred stocks, each class would receive a different ISIN. Similarly, multiple bond issues from the same company would have distinct ISINs, since each issue has unique specifications

such as the start date, end date, and interest rate. The same principle applies to more complex financial products, such as derivatives.

ISIN codes are issued and maintained by each country's National Numbering Agency or the Substitute Numbering Agency. The ISIN system has gained widespread market adoption and is increasingly recognized as the global standard for financial instrument identification. Due to their reliability, ISIN identifiers are predominantly used for trading, settlement, clearing, and regulatory reporting.

The 12 characters of an ISIN can be divided into three parts: the first two characters are the [ISO 3166-1 alpha-2 code](#) of the issuing country. The middle nine alphanumeric characters represent the security identifier. This nine-digit alphanumeric code is often called the National Securities Identifying Number (NSIN). Examples of NSINs include CUSIP in the USA, SEDOL in the UK, and WKN ("Wertpapierkennnummer") in Germany.⁵ The last character is a check digit computed using the modulus 10 double-add-double or Luhn algorithm. [Figure 3-7](#) offers a visual breakdown of the ISIN.

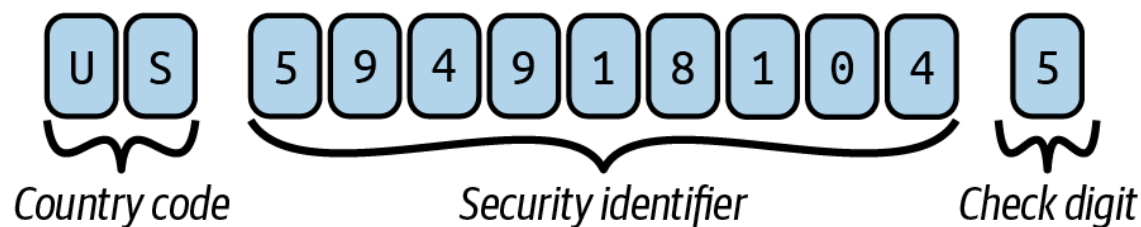


Figure 3-7. Structural breakdown of the ISIN code

One potential drawback of the ISIN is its international scope, which means it does not specify the trading location or currency. If a given stock trades on multiple different exchanges, the associated ISIN will be the same. For instance, IBM's common stock is listed on almost 25 trading platforms and exchanges around the world. In this case, some identifiers for IBM stock would vary depending on where it is traded (e.g., the ticker). However, IBM stock will have only one ISIN for each security.⁶ To overcome this limitation, the *Market Identifier Code* (MIC), defined in [ISO 10383](#), is commonly used alongside the ISIN to specify exchanges, trading venues, and both regulated and nonregulated markets.

Another limitation is that the ISIN does not provide detailed information about the contract terms or variations within a particular type of security. In other words, an ISIN cannot describe a contract uniquely.

An ISIN might become inactive or be replaced with a new one following corporate actions such as mergers and acquisitions, company name changes, stock splits, and the redemption/conversion of debt instruments.

Another limitation of the ISIN is that it does not cover all instruments, especially over-the-counter instruments. In this case, alternative instrument identifiers are often used.⁷

The Luhn algorithm or Luhn formula, also known as the modulus 10 or mod 10 algorithm, named after its creator, IBM scientist Hans Peter Luhn, and specified in [ISO/IEC 7812-1](#), is a widely used checksum formula for financial identifier validation. The algorithm is not meant to be a security measure against malicious attacks but rather a mechanism to protect against errors and distinguish valid identifiers from incorrect ones.

To validate an identifier such as ISIN using the Luhn formula, six steps are required:

1. If the identifier already contains the check digit, remove it. In most cases, the check digit is located at the end of the string. The remaining string constitutes the *payload*.
2. Starting from the rightmost digit in the payload, double the value of every second digit.
3. If doubling the digit results in a number that is greater than 9 (e.g., 18), then sum the digits of the doubled number to get a single digit (e.g., 12 becomes $1 + 2 = 3$).
4. Sum all the digits.
5. Compute the sum's modulus 10 as $(10 - (\text{sum} \bmod 10)) \bmod 10$. The result of this operation can be interpreted as the smallest number that needs to be added to the sum (possibly 0) to make it a multiple of 10.
6. If the modulus 10 obtained in step 5 is equal to the check digit, then the number is valid. Otherwise, it's not.

For an example, let's validate the ISIN of Microsoft, US5949181045:

1. To get a numerical string, we convert the country letters to digits by taking the ASCII code (American Standard Code for Information Interchange) of the capital letter and subtracting 55. The ASCII code for U is 85, while for S, it's 83; therefore, the new digit is $[85-55][83-55]5949181045 = 30285949181045$.
2. Remove the check digit (=5) to get the payload $\rightarrow 3028594918104$.
3. Starting from the right, we double the value of each second digit, and we get $[6] 0 [4] 8 [10] 9 [8] 9 [2] 8 [2] 0 [8]$.

4. Take the sum of the multidigit numbers (>9). We have one, which is [10], and by summing $1 + 0$, we get 1; therefore, the new sequence is [6] 0 [4] 8 [1] 9 [8] 9 [2] 8 [2] 0 [8].
5. Add up all the digits: $6 + 0 + 4 + 8 + 1 + 9 + 8 + 9 + 2 + 8 + 2 + 0 + 8 = 65$.
6. Calculate the check digit using the formula: $10 - (65 \bmod 10) \bmod 10 = 5$.
7. As the final result (5) is equal to the check digit, we can tell that the identifier is a valid ISIN.

A Python implementation for this ISIN check digit validation is available [in the GitHub repo for this book](#).

Classification of Financial Instruments

The *Classification of Financial Instruments* (CFI) is a six-letter code defined in [ISO 10962](#) that describes and classifies financial instruments. It was developed to address a major problem in financial markets, namely the need for a uniform and consistent approach to categorizing and grouping financial instruments. The ISO has appointed SIX Group, the national numbering agency of Switzerland, as the maintenance agency for the CFI code. SIX publishes a list of all CFI codes and modifications on [its website](#).

Since July 1, 2017, CFI codes are globally assigned alongside the ISIN when a new financial instrument is issued. In most cases, an instrument's CFI remains unmodified during its lifespan. However, corporate-related events such as changes to voting rights or ownership restrictions could cause a CFI to change.

In a CFI code, the first letter indicates the instrument category, the second denotes the subcategory, and the remaining letters indicate various attributes of the instrument. Both the second and remaining letters are optional. For example, the CFI code ESVXXX is for equity (E), common/ordinary share (S) with voting right (V), while code FXXXXX indicates a future (F), and OCXXXX is an option (O) of type call (C). [Figure 3-8](#) offers a visual illustration of the CFI code.

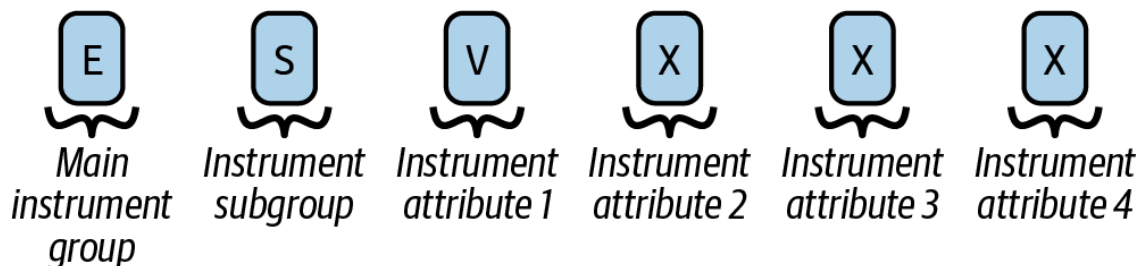


Figure 3-8. Structural breakdown of the CFI code

Financial Instrument Short Name

The *Financial Instrument Short Name* (FISN), defined in [ISO 18774](#), is a human-readable code used to provide a consistent, uniform, and global description of financial instruments. Since July 1, 2017, the FISN has been globally assigned alongside the ISIN and CFI at the time of issuance of a new financial instrument.

Unlike ISINs, which are primarily used for clearing and settlement, FISNs are utilized by market participants to enhance the readability, efficiency, reliability, data consistency, and transparency of financial services transactions and reference data.

The FISN code has a maximum length of 35 alphanumeric characters. Of these, 15 are reserved for the issuer name, and 19 for the description of the instrument, with the character “/” as the separator between the issuer name and description. [Figure 3-9](#) illustrates the structure of the FISN code.

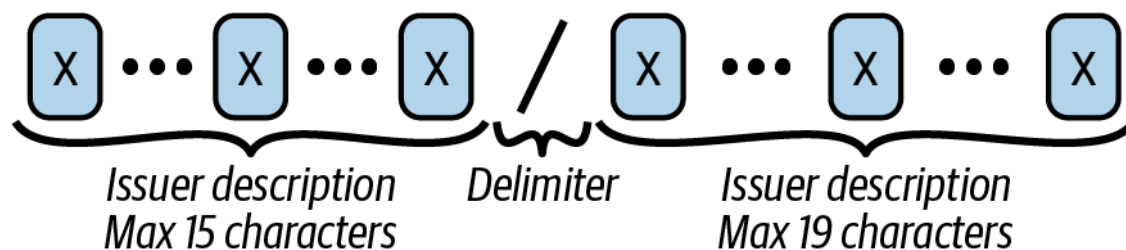


Figure 3-9. Structural breakdown of the FISN code

Committee on Uniform Security Identification Procedures

The *Committee on Uniform Security Identification Procedures* (CUSIP) is a nine-character alphanumeric code used to identify financial securities in the United States and Canada. CUSIP codes are mainly used for trading, settlement, and clearing. They are issued and managed by [CUSIP Global Services \(CGS\)](#), which acts as the US national numbering agency and is operated by FactSet Research Systems, Inc., as of the time this book was written.

The first six characters of a CUSIP uniquely identify an issuer (company, municipality, agency). The seventh and eighth characters identify the instrument type and issue using a hierarchical alphanumeric convention. The last character is a check digit.

For US securities, the CUSIP number makes up the middle nine characters of a US ISIN. For example, Amazon's ISIN is US0231351067, and Amazon's CUSIP is 023135106. [Figure 3-10](#) offers a visual representation of the CUSIP code.

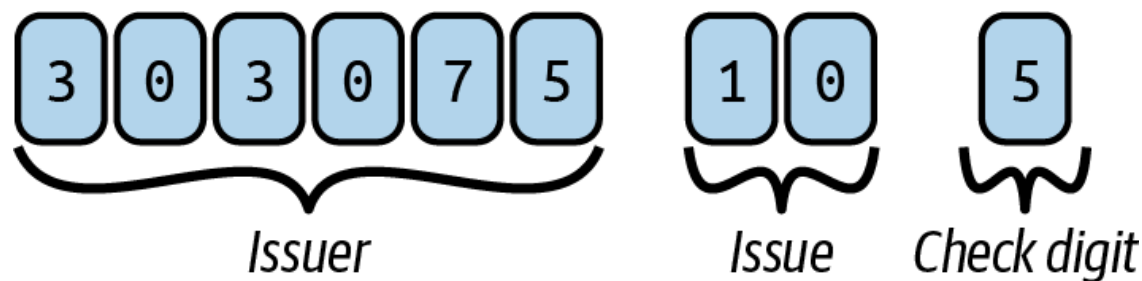


Figure 3-10. Structural breakdown of the CUSIP code

To check the validity of a CUSIP code, the following steps are required:

1. Convert non-numeric characters to digits according to their ordinal position in the alphabet plus 9 (e.g., A = (1 + 9) = 10).
2. Convert the characters * to 36, @ to 37, and # to 38.
3. Multiply every even digit by 2. If the result is a two-digit number, add the digits together (e.g., 13 → 1 + 3 = 4).
4. Get the sum of all values.
5. Get the floored value of the sum: $(10 - (\text{sum modulo } 10)) \text{ modulo } 10$.

A Python implementation for ISIN check digit validation is available [in the GitHub repo for this book](#).

Recently, a new identifier, called *CUSIP Entity Identifier* (CEI), was [introduced by CGS](#) to identify legal parties involved in the syndicated lending market.⁸ The CEI was developed in collaboration with the Loan Syndications and Trading Association (LSTA) and syndicated loan solution providers.

Legal Entity Identifier

The *Legal Entity Identifier* (LEI), defined in [ISO 17442](#), is a 20-character alphanumeric code used to identify legal entities engaged in financial transactions. Each LEI contains information about an entity's ownership structure and thus answers the questions of “who is who” and “who owns whom.” This concept aligns closely with know your customer (KYC) practices in financial markets, which aim to verify the identity of clients within financial institutions. The LEI identifier is a global, unique, and freely accessible identifier. Its official maintainer is the Global Legal Entity Identifier Foundation (GLEIF).

As illustrated in [Figure 3-11](#), the first four characters of the LEI form a prefix that identifies the Local Operating Unit (LOU) that issued the LEI. Characters 5 to 18 constitute the entity identifier assigned by the LOU. Finally, characters 19 and 20 are two check digits.

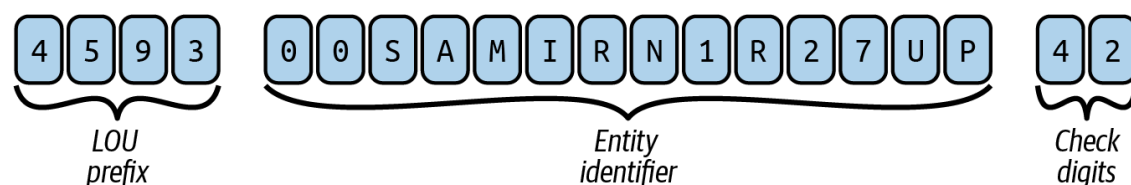


Figure 3-11. Structural breakdown of the LEI identifier

ISO 17442 [classifies](#) the following as legal entities:

- All financial intermediaries
- Banks and finance companies
- International branches

- All entities that issue equity, debt, or other securities for other capital structures
- All entities listed on an exchange
- All entities that trade financial instruments or are otherwise parties to financial transactions, including business entities, pension funds, and investment vehicles such as collective investment funds (at umbrella and subfund levels) and other special purpose vehicles that have a legal form
- All entities under the purview of a financial regulator and their affiliates, subsidiaries, and holding companies
- Sole traders (as an example of individuals acting in a business capacity)

On April 4th, 2019, ANNA and GLEIF launched a collaborative initiative to associate ISIN identifiers with the LEI identifiers of their issuers. This initiative aims to enhance market transparency by establishing a direct link between the issuer and the issuance of securities.

The introduction of the LEI has been celebrated as a major milestone in financial markets, generating numerous discussions and publications about its beneficial impact on improving market efficiency and stability. The question is, why all this attention?

The issue started shortly after the collapse of Lehman Brothers in 2008. As explained by a [McKinsey report](#), “After Lehman’s demise, participants in the global financial system could not assess their exposure to Lehman, its subsidiaries, and each other because there was no standard system for identifying counterparties in the maze of subsidiaries and affiliates from which banks, insurers, asset managers, and other market participants transact.”

To give an illustrative example, when Lehman Brothers collapsed in 2008, it had an [estimated 8,000 subsidiaries working in various jurisdictions](#), while Morgan Stanley had roughly 3,500 subsidiaries. Consider the potential number of bilateral agreements and transactions between any of Lehman’s 8,000 subsidiaries and Morgan Stanley’s 3,500 and assume that such agreements are specified using different identification systems. Now, assume that you want to calculate the total exposure of Morgan Stanley to Lehman Brothers. This would require aggregating millions of positions, each identified using the subsidiary’s convention.

Unsurprisingly, it took quite a lot of time for Lehman Brothers’s counterparties to calculate their total exposure to the consolidated entity. To mitigate this issue and ensure that financial institutions can quickly aggregate their exposures and risk information, the ISO led an initiative to develop an international legal entity identification standard, resulting in the LEI.

In addition to the LEI, the ISO created the *Entity Legal Forms* (ELF), described in [ISO 20275](#), to uniquely identify entity forms/types globally using a standardized four-character code. GLEIF maintains ELF codes that can be accessed and downloaded from its [web page](#).

Transaction Identifiers

Identifiers are the backbone of the lifecycle of financial transactions. During its processing, a single transaction often traverses various systems and parties. Consequently, the same transaction might be recorded with different IDs as it moves from one system to the next. This makes it difficult to unambiguously identify transactions and all related messages across disparate systems and databases. To overcome this issue, unique transaction identifiers have been developed.

One example is the *Unique Swap Identifier* (USI),⁹ specific to the United States and mandated by the Commodity Futures Trading Commission (CFTC) and the Securities and Exchange Commission (SEC) as part of the Dodd-Frank Act. The USI is a fixed-length identifier assigned to all swap transactions, identifying the transaction uniquely throughout its lifecycle. To expand the scope of the USI identifier for global reporting of financial transactions, the *Unique Trade Identifier* (UTI) was introduced.

To ensure consistency across different jurisdictions and reporting platforms, the ISO introduced the ISO 23897—the *Unique Transaction Identifier* (UTI) standard. It provides specifications aimed at standardizing transaction identifiers globally, ensuring unique identification throughout a financial transaction's lifecycle. This is essential for facilitating transaction reporting, improving traceability, and reducing operational errors, especially as cross-border trading expands.¹⁰

The format of these transaction identifiers is generally similar. ISO 23897 specifies a max length of 52 characters for UTI code, but a variable length is possible. There is a consensus that a unique transaction identifier should include a prefix identifying the issuing entity, followed by a string that uniquely identifies the transaction. A market preference has emerged in favor of using international standards such as LEI identifiers for the prefix part.¹¹ For example, in 2013, the ISDA working group proposed a best practice recommendation for the UTI, where the prefix consists of characters 7–16 of the LEI, followed by a 32-character string, as illustrated in Figure 3-12.

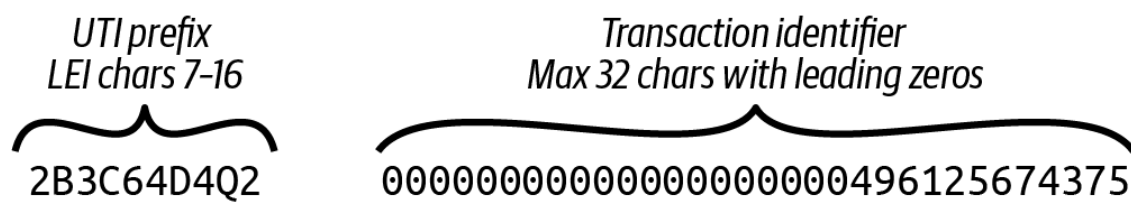


Figure 3-12. ISDA-based Unique Trade Identifier (UTI) structure

Stock Exchange Daily Official List

The *Stock Exchange Daily Official List* (SEDOL) is a seven-character alphanumeric code mainly used to identify securities traded on the London Stock Exchange (LSE) and other smaller exchanges in the UK. The LSE (the National Numbering Agency of the UK) assigns SEDOL codes upon request from the security issuer.¹²

Over the years, the SEDOL system has expanded, and SEDOL codes can now be issued at the country level to represent securities listed in multiple jurisdictions. SEDOLs are issued globally across all jurisdictions and multiple asset classes. If a security is traded on a different exchange in a different country, it will be assigned a separate SEDOL code. This makes SEDOLs unique across countries, which is ideal for international trading and security identification.

For UK securities, SEDOLs are embedded within the UK ISIN codes by adding the country code at the beginning, followed by two padding zeros, then the SEDOL, and finally, the ISIN check code. For example, the UK banking group HSBC has a SEDOL code of 0540528 and an ISIN code of GB0005405286.¹³

SEDOL codes issued prior to March 2004 were exclusively numeric. Afterward, the SEDOL system moved to an alphanumeric format that starts with an alphabetic character, followed by five alphanumeric characters and a trailing numeric check digit. [Figure 3-13](#) illustrates the structure of SEDOL.

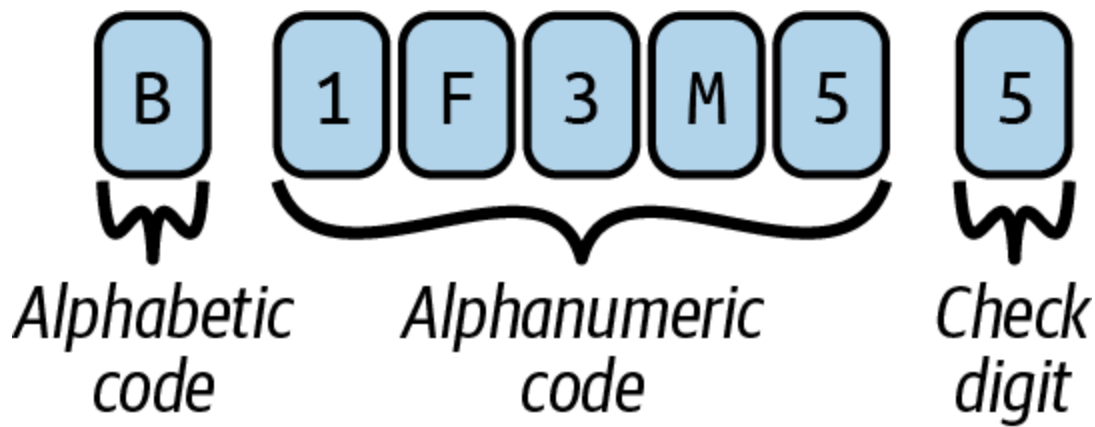


Figure 3-13. The structural breakdown of the SEDOL identifier

To check if a SEDOL is valid, the check digit is chosen to make the weighted sum of all SEDOL characters a multiple of 10. The steps to validate a SEDOL are as follows:

1. Convert non-numeric characters to digits according to their ordinal position in the alphabet plus 9 ($A = (1 + 9) = 10$).
2. Multiply each of the first six numbers by their corresponding weight: 1 for first position, 3 for second, 1 for third, 7 for fourth, 3 for fifth, 9 for sixth.
3. Get the sum of all values.
4. Get the floored value of sum: $(10 - (\text{sum modulo } 10)) \text{ modulo } 10$.

A Python implementation for SEDOL check digit validation is available [in the GitHub repo for this book](#).

Ticker Symbols

A ticker symbol is a short and unique series of letters assigned to financial securities (mostly stocks) for listing and trading purposes. There is no standard for tickers, and they can be generated and assigned by various organizations, including exchanges and trading venues, financial data providers, and financial institutions.

The allocation process and formatting conventions of tickers are specific to each issuing organization. Among US exchanges, tickers are typically one to four characters long, and they resemble the company name when possible. For example, the ticker for Apple, Inc. on the NYSE is AAPL,

while Ford Motor has the ticker F. Some exchanges, such as NASDAQ, [add a fifth symbol to their tickers](#) to convey information about the trading status and special features of the stock. For example, in BRK.A, the first three represent the stock symbol for Berkshire Hathaway, Inc. (BRK), and the last letter (.A) indicates that the shares are of class A type, which traditionally holds more voting rights.

In addition to exchanges, financial data providers assign proprietary ticker symbols to financial instruments. For example, Bloomberg created its own Bloomberg ticker to identify a financial entity uniquely within the Bloomberg ecosystem. A Bloomberg ticker can include the exchange-specific ticker, the market sector, the exchange code, instrument-specific information (e.g., bond maturity, option expiry, option type, etc.), and the Bloomberg database (e.g., EQUITY for stocks).

Another well-known proprietary ticker symbol is the *Refinitiv Instrument Code* (RIC), which is issued and maintained by LSEG. RIC tickers are mainly used to look up information on specific financial instruments on LSEG platforms. The main component of the RIC is the security's ticker symbol with an optional character that identifies the exchange. For example, the RIC symbol IBM.N refers to IBM stock (IBM) traded on the New York Stock Exchange (.N).

Stock tickers can vary by exchange and country, which implies that they may not be unique across different exchanges and countries. As a result, to reliably identify a stock, both the ticker and the exchange or country of listing are often required.

Tickers are not immutable and might change to reflect corporate actions such as mergers and acquisitions. For example, prior to the 1999 merger with Mobil Oil, Exxon used the phonetic spelling of the company (XON) as its ticker symbol. After the merger, the symbol changed to XOM.

Importantly, tickers are not guaranteed to remain unique, as they can be reassigned over time. For example, until 2017, the ticker SNOW was assigned by NYSE to Intrawest Resorts Holdings, Inc. In May 2017, Henry Crown and Company and KSL Capital Partners acquired Intrawest and transformed it into a privately owned company. After delisting Intrawest,

the ticker SNOW was reassigned to Snowflake. A Wikipedia search of both company names would confirm this, as illustrated in [Figure 3-14](#).

			
Type	Private	Type	Public company
Traded as	NYSE: SNOW	Traded as	NYSE: SNOW ↗ (Class A) Russell 1000 component

Figure 3-14. NYSE tickers for Intrawest and Snowflake

Derivative Identifiers

Derivatives are among the most exchanged financial securities in financial markets. As a quick reminder, a derivative is a contract that derives its value from an underlying financial asset or variable such as stocks, commodities, foreign exchange, interest rates, indexes, and many more.

Crucially, identifying derivative instruments is more challenging than identifying other types of financial instruments. First, several constituent elements must be considered to identify a derivative instrument. Second, due to their flexible and customizable nature, derivatives can easily turn into very complex products. Third, a substantial deal of derivatives is traded OTC, complicating their tracking and identification. Nonetheless, as I will demonstrate next, market participants have developed various initiatives to identify both exchange-traded and OTC derivative instruments.

Option symbol

Option symbols are derived symbols used to identify an option and its characteristics on a given exchange.¹⁴ The current market standard for option symbols is based on the *Options Clearing Corporation's* (OCC) *Options Symbolology Initiative* (OSI). The OSI format consists of a 21-character alphanumeric code that can be split into four parts:

- A root (ticker) symbol of the underlying stock or ETF (exchange-traded fund).
- An expiration date, which is six digits in the format YYMMDD.

- An option type, either C for a call or P for put.
- The strike price. This is represented by the price times 1,000, with the front padded with 0s to 8 digits. The decimal point falls three places from the right in the options symbol: 00000.000.

The structure of the OSI option symbol is illustrated in detail in [Figure 3-15](#).

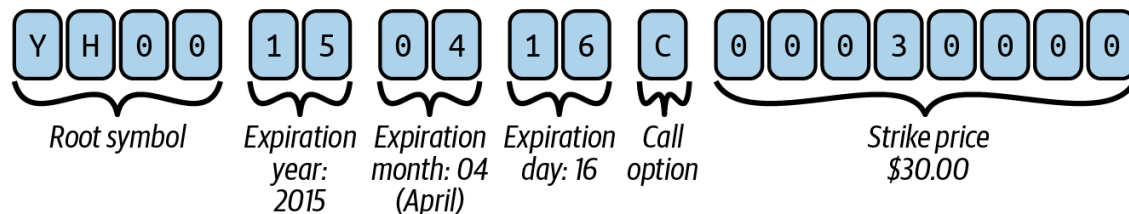


Figure 3-15. The structural breakdown of the OSI option symbol

CFI, UPI, and OTC ISIN

To identify OTC derivatives, a combined identification scheme has been put in place that relies on three identifiers:

OTC ISIN

This is allocated by the *Derivatives Service Bureau* (DSB), with the initial two characters starting with the custom “EZ” code.

Unique Product Identifier (UPI)

This is defined in [ISO 4914](#) for the identification of OTC derivative products.

Classification of financial instruments (CFI)

This is generated by the DSB as part of the OTC ISIN generation process.

The three identifiers are combined to provide different levels of detail. Malavika Solanki, of the management team at the DSB, [illustrated the combined use of the three identifiers with the following example](#): at the highest level, the CFI can tell that a derivative instrument is a “single currency, fix-float, interest rate swap with a constant notional schedule and cash delivery.” The UPI would tell a bit more about the product, for exam-

ple, that it has a “three-month reference rate term, a USD reference rate, and that the name of the reference rate was USD-LIBOR-BBAR.” Finally, the OTC ISIN can provide more granular details about the specific instrument that has been transacted, such as “the standardized ISO reference rate name, the price multiplier associated with the instrument, the full name and short names of the instrument, the expiry date.”

Alternative Instrument Identifier

The *Alternative Instrument Identifier* (AII) has been adopted within the European Union for reporting purposes to identify derivatives traded on regulated markets without an ISIN assigned to them. Rather than being a code, the AII is a concatenation of descriptive fields that identify the instrument. The fields include the exchange code, exchange product code, derivative type, put/call identifier, expiry/delivery/prompt date, and strike price, as illustrated in [Figure 3-16](#).

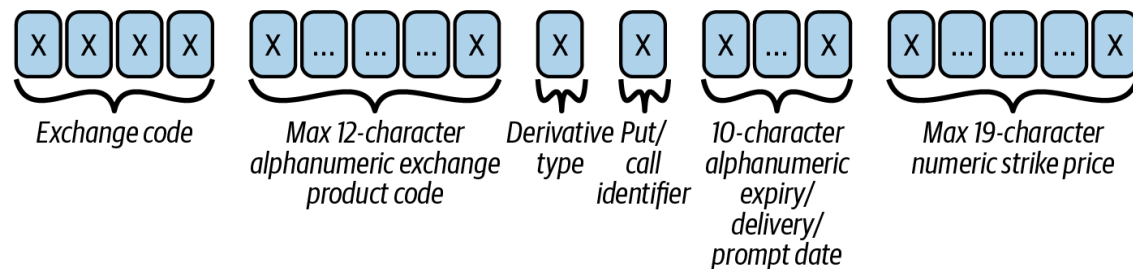


Figure 3-16. The structural breakdown of the AII identifier

Financial Instrument Global Identifier

The *Financial Instrument Global Identifier* (FIGI) is a 12-character alphanumeric ID covering hundreds of millions of active and inactive instruments around the world.

The history of FIGI started in 2009 when Bloomberg decided to release its Open Symbology (BSYM) system for identifying financial securities across asset classes. The BSYM provides a library of identifiers for hundreds of millions of securities, known as Bloomberg Global Identifiers (BBGIDs). In 2014, the BSYM symbology system was adopted by the Object Management Group (OMG), a nonprofit standards consortium, in order to promote it as an open industry standard. Subsequently, the BBGID was re-

named to FIGI. Since its introduction, FIGI has received a lot of market attention. For example, the Accredited Standards Committee X9 [adopted FIGI as an official US data identification standard.](#)

FIGI codes are unique across all markets and countries and remain unchanged once issued. For example, if IBM stock is traded on 12 stock exchanges, there will be 12 different FIGIs.

Furthermore, FIGI was designed as a global system capable of identifying any type of financial instrument, including stocks, bonds, derivatives, loans, indexes, funds, and digital assets. FIGIs are also free to access. An open source tool called [OpenFIGI](#) was released to identify, map, and request a free FIGI via an API.

The FIGI identification system has a hierarchical structure composed of three levels:

Global FIGI

This is the most granular level, identifying a financial asset at the trading-venue level. It is unique to a specific instrument at a particular trading venue.

Composite global FIGI

This level aggregates multiple venue-level FIGI identifiers within the same country, providing a broader identification that encompasses all trading venues for a specific instrument within that country.

Share class global FIGI

This level further aggregates FIGI identifiers to cover financial instruments across multiple countries. It provides a global view of a single instrument regardless of the country and venue.

For example, Amazon trades on multiple stock exchanges in the US. When Amazon common stock trades on the New York Stock Exchange, it has the FIGI code BBG000BVPXP1. When the same stock trades on the NASDAQ Global Select exchange, it has a FIGI of BBG000BVQ4Z3. But if

you don't care about which US exchange Amazon trades on, you can use the Amazon composite FIGI (BBG000BVPV84) to generically reference Amazon stock traded in the US. If you want to globally identify Amazon common stock regardless of trading venue and country, the share class code (BBG001S5PQL7) can be used.¹⁵ [Figure 3-17](#) illustrates the hierarchical structure of FIGI.

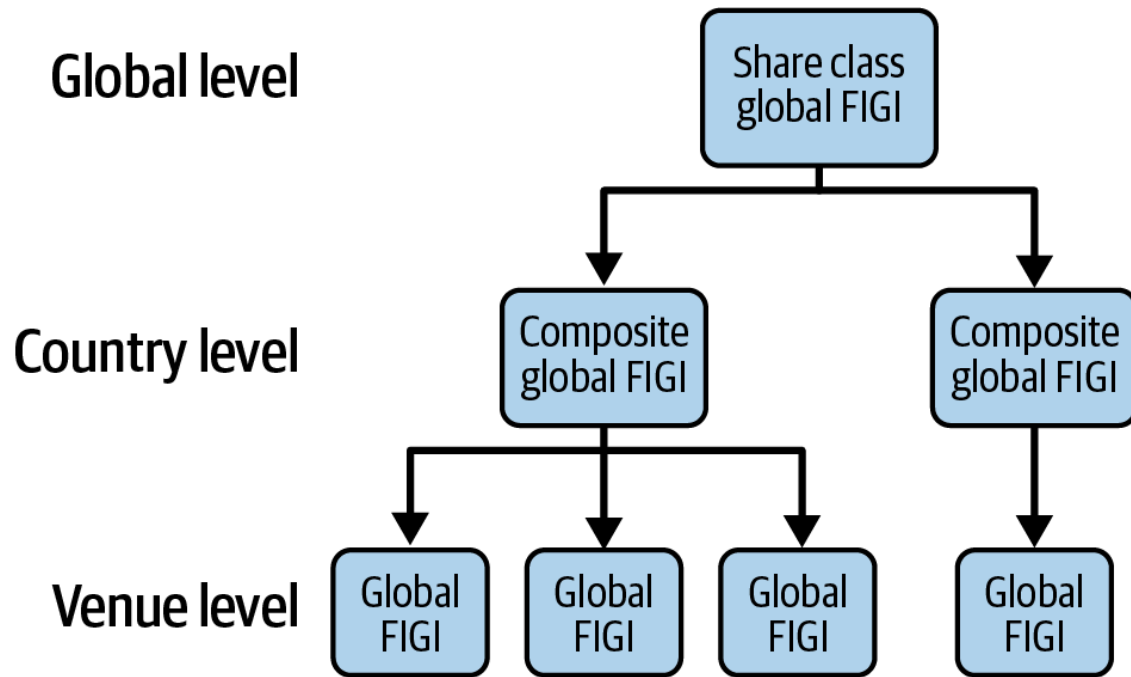


Figure 3-17. Hierarchical representations of FIGI

All three levels of FIGI identifiers have the same structure and limitations. The identifier may contain only letters in [B, C, D, F, G, H, J, K, L, M, N, P, Q, R, S, T, V, W, X, Y, Z], and zero to nine digits.

The structure of FIGI codes is as follows:

- The first two characters identify the certified issuer that created the FIGI code. Currently, Bloomberg generates the majority of FIGI codes, hence the presence of the prefix BB in most identifiers.
- The third character is the letter G, used to indicate that it's a global identifier.
- Characters 4–11 are alphanumeric characters that constitute the reference ID.
- A trailing check digit.

[Figure 3-18](#) illustrates this structure visually.

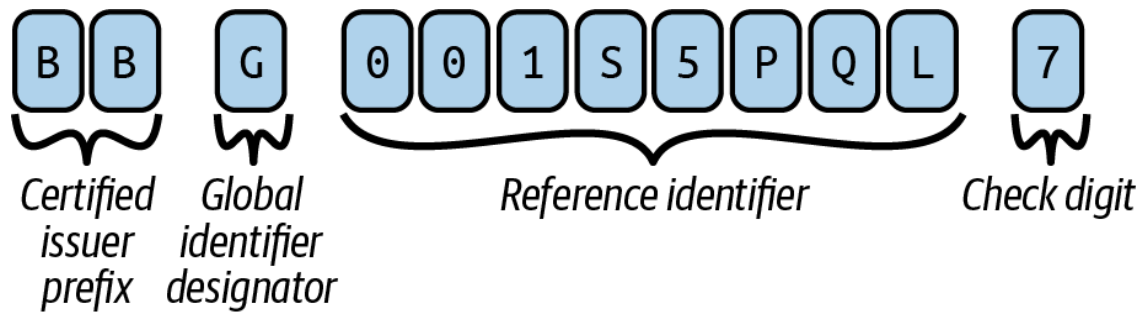


Figure 3-18. The structural breakdown of FIGI

The check digit procedure is based on the Luhn algorithm, following five steps:

1. Get the identifier, e.g., BBG000BLNQ16, and remove the check digit, BBG000BLNQ1.
2. Convert non-numeric characters to digits according to their ordinal position in the alphabet plus 9 (A = (1 + 9) = 10). In our example, we get [11][11][16][0][0][0][11][21][23][26][1].
3. Double every second digit: [11][22][16][0][0][0][11][42][23][52][1].
4. Compute the sum of the resulting values: $1 + 1 + 2 + 2 + 1 + 6 + 0 + 0 + 0 + 1 = 34$.
5. Get the floored value of the sum: $(10 - (34 \text{ modulo } 10)) \text{ modulo } 10 = 6$, which is the check digit attached to the identifier.

A Python implementation for FIGI check digit validation is available in this book's [GitHub repo](#).

FactSet Permanent Identifier

The *FactSet permanent identifier* is a proprietary identification system developed by FactSet to offer a stable and unified identifier. It includes three levels:

Security

Identifying the security globally

Regional

Identifying the security at the regional level per currency (e.g., US/USD)

Listing

Identifying the security at the market level (e.g., US/NYSE/USD)

FactSet provides the [FactSet Symbolology API](#), a symbol/identifier resolution service that allows users to map a wide variety of identifiers to FactSet's native symbology or third-party identifiers such as CUSIPs, SEDOLS, ISINs, Bloomberg FIGI, and many more. In addition, FactSet offers the [FactSet Concordance API](#), which enables users to programmatically match the FactSet identifier for a specific entity based on attributes, such as name, URL, and location.

LSEG Permanent Identifier

The [*Permanent Identifier \(PermID\)*](#) is a unique identifier used within the London Stock Exchange Group (LSEG) information model to identify and reference various objects, including organizations, instruments, funds, issuers, and people. It ensures that these objects are accurately and unambiguously referenced and linked together, even as their relationships change over time.

What makes PermID especially valuable is that it's linked to a unique web address or Uniform Resource Identifier (URI), offering a permanent, direct link to the identified entity. LSEG has made PermID open source, enabling access via web pages or through API-based entity search and matching services.

For example, the following PermID URLs can be used to retrieve information about the technology company Apple:

[*https://permid.org/1-4295905573*](https://permid.org/1-4295905573)

Provides organizational details identifying Apple, Inc.

[*https://permid.org/1-8590932301*](https://permid.org/1-8590932301)

Contains instrument details identifying Apple's ordinary shares.

[*https://permid.org/1-25727408109*](https://permid.org/1-25727408109)

Includes quote information identifying Apple's ordinary shares on the New York Stock Exchange.

Digital Asset Identifiers

Following the emergence and diffusion of blockchain and distributed ledger technologies, *digital assets* have been established as a new type of financial market entity. According to the [ISO 22739 vocabulary for blockchain](#), a digital asset is an “asset that exists only in digital form or which is the digital representation of another asset.” In many cases, digital assets are referred to as *digital tokens*. A digital token can be *fungible* if it's identical and interchangeable with similar assets (e.g., one dollar in New York is the same as one dollar in Australia), or *nonfungible* if they are unique and nondivisible (e.g., a painting or a boat). The most common place for exchanging digital assets is a *blockchain*, which, according to ISO 22739 vocabulary, refers to a “distributed ledger with confirmed blocks organized in an append-only, sequential chain using cryptographic links.”¹⁶ The process of converting something to a digital asset and adding it to a blockchain system is called *tokenization*.

Over the past years, a wide variety of fungible digital assets have been developed, for example the following:

Crypto assets

Assets developed using cryptographic techniques. The most notable examples are cryptocurrencies such as Bitcoin and Ethereum.

Crypto assets can serve as a means of payment or investment.

Security token

A financial security or instrument converted into a digital token and exchanged on a distributed ledger like blockchain (e.g., Microsoft Stock Token). Similar to traditional security certificates, security tokens represent an ownership right in a company or asset.

Nonfungible tokens

These represent ownership rights over a unique digital asset such as images, videos, and games.

Utility token

A token used to access a specific service or feature within a blockchain-based ecosystem.

As digital assets have grown in popularity, the requirement to identify the issued, stored, and transacted digital tokens and crypto assets has become more urgent. A first initiative was established with the introduction of the *Digital Token Identifier* (DTI), defined in [ISO 24165-1](#), to identify fungible tokens and digital ledgers. In defining the DTI, the ISO realized the need for a new methodology for assigning the identifier. This is because digital assets are often not associated with a particular issuing entity, such as central banks, unlike traditional assets.

The standard helps in reducing confusion and increasing trust in the crypto assets market by providing a universal method of identification.

Industry and Sector Identifiers

Industry and sector identifiers are classification frameworks used to categorize and group businesses into various categories and subcategories. Financial firms use industry identifiers to analyze market exposure, classify stocks, measure concentration risks, and understand cross-industry differences. Two main industry classification frameworks are widely used: the *Standard Industrial Classification* (SIC) and the *North American Industry Classification System* (NAICS).

The SIC identifier is a four-character numerical code used to classify industries based on their primary activities. The SIC system was created in 1937 by the US government to facilitate economic analysis across industries and agencies and to promote standardization and uniformity in the collection and recording of industrial data.

As illustrated in [Figure 3-19](#), the official SIC classification code consists of three parts. The first two digits, which are mandatory, identify the major

sector group. The third digit further categorizes the business into an industry group. The fourth digit provides the most granular classification, specifying the particular sector of the business. For example, [SIC code 6021 \(National Commercial Banks\)](#) belongs to industry group 602 (Commercial Banks), which is part of major group 60 (Depository Institutions), which belongs to the division of (Finance, Insurance, and Real Estate).

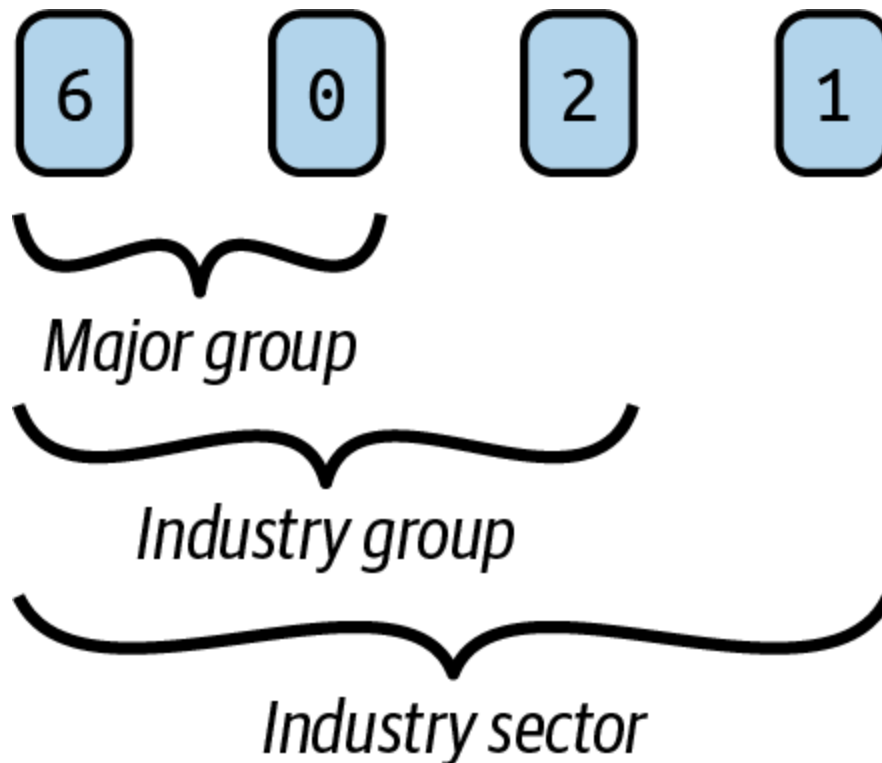


Figure 3-19. The structural breakdown of the SIC code

The SIC system had several shortcomings. First, it produced ambiguous, mismatched, and overlapping classifications generated by SIC. Second, the four-digit system restricted the addition of new, emerging business sectors and industries. To overcome these issues, the SIC system was replaced in 1997 by the North American Industrial Classification System (NAICS), which introduced a more flexible six-character numeric code.

As shown in [Figure 3-20](#), the first two digits of an NAICS code indicate the major sector of the business. The third digit indicates the subsector, and the fourth digit designates the industry group. The fifth digit indicates the specific industry of operation, while the sixth code specifies the national industry. For example, [NAICS code 522110](#) is used to identify commercial banks. The first two digits, 52, define the sector (Finance and Insurance),

the first three digits (522) identify the subsector (Credit Intermediation and Related Activities), the first four digits (5221) define the industry group (Depository Credit Intermediation), and the last two digits identify the industry (Commercial Banking).

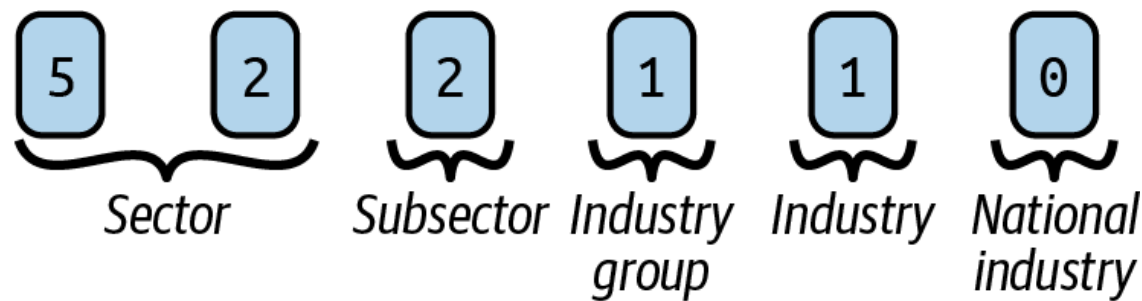


Figure 3-20. The structural breakdown of the NAICS code

Other industrial classification frameworks are in use worldwide. These include the Statistical Classification of Economic Activities in the European Community, known as NACE and predominantly used in the European Union, the UK Standard Industrial Classification of Economic Activities (UK SIC), and the Australian and New Zealand Standard Industrial Classification (ANZSIC).

Bank Identifiers

Banks are the cornerstones of financial markets. They provide a secure and reliable place for individuals and organizations to deposit, transfer, and invest money, obtain loans, and make credit and debit card payments. Additionally, they support various online applications and offer a platform for FinTech firms to offer their services. Therefore, bank identifiers are essential for identifying banks, customers, and payment cards.

The most prominent example is the *Business Identifier Code* (BIC), also called SWIFT or Bank Identifier Code. This alphanumeric code, defined in [ISO 9362](#), is used to identify banks, financial institutions, and nonfinancial institutions worldwide when conducting international money transfers and routing exchanging messages. SWIFT issues BIC codes.

A SWIFT/BIC consists of either 8 or 11 alphanumeric characters that identify the country, city, bank, and optionally the branch:

Bank code

Four-character alphabetic characters identifying the bank. It usually looks like a shortened version of that bank's name.

Country code

A two-character ISO 3166-1 alpha-2 code indicating the country where the bank is located.

Location code

Two-character alphanumeric code that designates where the bank's main office is.

Branch code

An optional three-character alphanumeric code representing a specific branch. XXX is used to indicate the bank's head office.

Figure 3-21 illustrates the structure of the BIC. The example shown refers to the Italian bank UniCredit Banca. The displayed code can be read as follows: UNCR identifies UniCredit Banca, IT is the country code for Italy, MM is the office location code for Milan, and XXX indicates the head office.

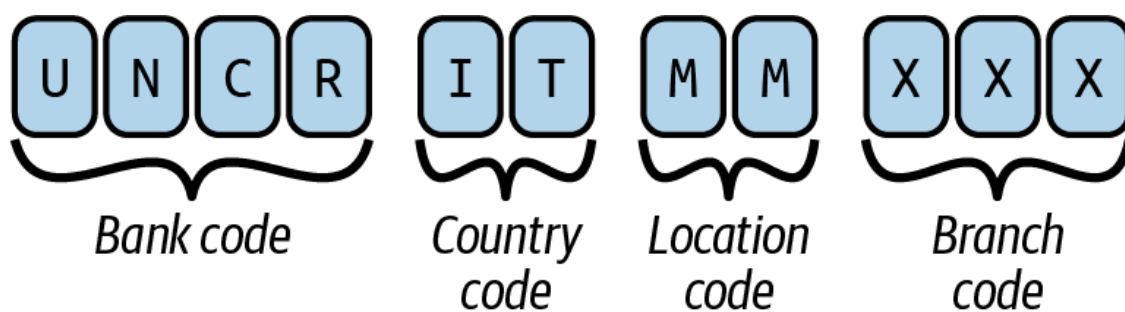


Figure 3-21. The structural breakdown of the BIC code

The use of the BIC code to identify financial institutions is quite common, especially when identifying financial institutions in international transactions. Nevertheless, certain countries employ their own local bank or branch identification codes. For instance, Australia uses the *Bank State Branch* (BSB), a six-digit code, to identify branches of Australian financial institutions. Similarly, in the United States, financial institutions involved

in various payment operations are identified using the *ABA Routing Number*, consisting of nine digits.

Another well-known bank identifier is the *International Bank Account Number* (IBAN). It is defined in [ISO 13616](#) and serves as an international system of unique codes used to identify bank accounts when conducting money transfers. The IBAN system was originally developed for use within the EU, but it was later adopted by regions such as the Caribbean and Middle East.

The length of the IBAN varies by country, but it cannot exceed 34 characters. For instance, Belgian IBANs have 16 alphanumeric characters, Luxembourg's are 20, and Germany's are 22.

An IBAN code is composed of three main parts (see [Figure 3-22](#) for an illustration):

- Country code following the ISO 3166-1 alpha-2 convention.
- Two check digits.
- *Basic Bank Account Number* (BBAN)—up to 30 alphanumeric characters that include the bank code, branch identifier, and account number. The length of BBAN may vary across countries.

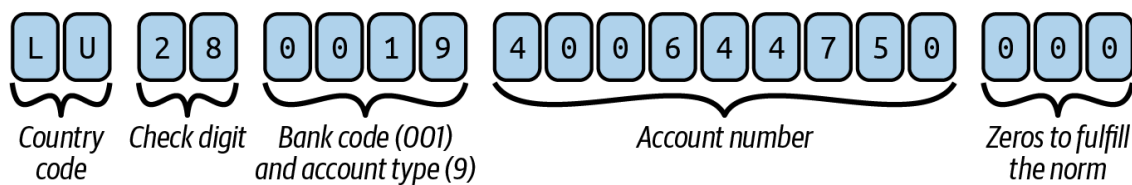


Figure 3-22. The structural breakdown of the IBAN number for Luxembourg

An IBAN code is validated through a mod-97 algorithm (as described by [ISO 7064](#)) that works as follows:

1. Make sure the IBAN has a valid length. For example, let's take a random Luxembourgish IBAN: LU280019400644750000. The length is 20, which matches the country's specified length.
2. Move the first four characters to the end of the string → 0019400644750000LU28.

3. Convert non-numeric characters to digits according to their ordinal position in the alphabet plus 9 ($A = (1 + 9) = 10$). In our example, we get 0019400644750000213028.
4. Treat the result as an integer and check if the modulo 97 of the number is equal to 1, $19400644750000213028 \bmod 97 = 1$.

A Python implementation for IBAN validation is available in the [GitHub repo for this book](#).

Last but not least, an important bank identifier is the payment card number or *Primary Account Number* (PAN), defined in [ISO/IEC 7812](#). This is used to define payment cards and identify their issuer and cardholder. Most payment cards, such as credit, debt, and gift cards, have their PAN laser-printed on the front (yes, it's your credit card number!).

The PAN is a numeric code with variable length ranging from 8 to 19 digits. The first six to eight digits represent the *Issuer Identification Number* (IIN), which identifies the card issuer. The first digit in the IIN is the *Major Industry Identifier* (MII), which identifies the industry/sector of the card issuer (for example, 4 and 5 are commonly used to identify financial institutions). The remaining characters are the individual account numbers used to identify the cardholder's account. The last digit is a check digit, which can be validated using the Luhn algorithm.¹⁷ [Figure 3-23](#) illustrates the structure of the PAN.

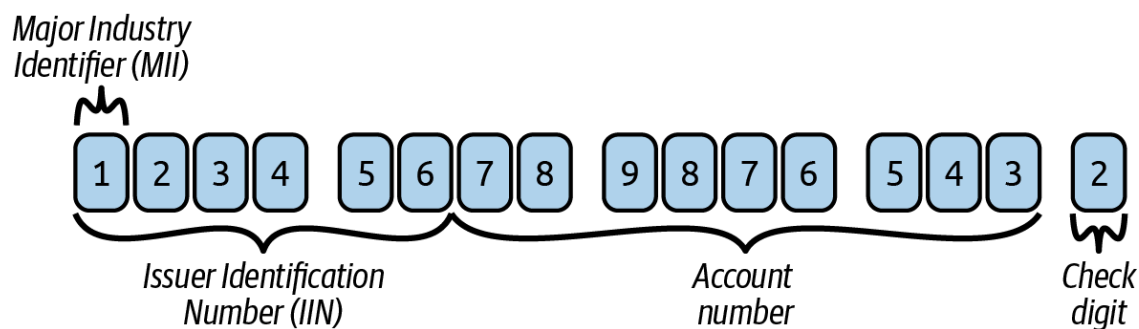


Figure 3-23. The structural breakdown of a 16-digit PAN Identifier

Summary

This chapter offered an in-depth treatment of financial identifiers and identification systems, which can be summarized as follows:

- Defining financial identifiers and identification systems and highlighting their critical role in financial market operations.
- Providing an overview of the entities involved in the creation, issuance, and maintenance of financial identifiers and identification systems.
- Outlining the desired properties of an optimal financial identification system.
- Discussing existing financial identification systems in detail, highlighting their main features and shortcomings.

The key takeaway from this chapter is that financial identification presents significant challenges for financial markets. In addition, as new products and market requirements emerge, the complexity of the issue continues to evolve. I highly recommend you stay informed about emerging trends in the development, standardization, and adoption of financial identification systems, as well as the evolving characteristics of existing systems.

Now that you have a solid understanding of financial identification systems, let's move on to explore another related and crucial problem in financial markets: financial entity systems. You build a financial entity system when you want to extract, recognize, identify, and match financial entities within and across various sources of financial data.

The next chapter is all about financial entity systems—let's keep moving!

1 If you are interested in this topic, I highly recommend Martijn Groot's book, *Managing Financial Information in the Trade Lifecycle: A Concise Atlas of Financial Instruments and Processes* (Elsevier, 2008).

- ² For an interesting read, see Richard Young's article, ["The Identifier Challenge: Attributes of MiFID II That Cannot Be Ignored"](#), *Journal of Securities Operations & Custody* 9, no. 4 (Autumn 2017): 313–320.
- ³ In 2011, the European Commission [concluded](#) that Standard and Poor's (S&P) was abusing its dominant position and violating antitrust rules as the unique issuing agency of US security ISINs by charging European financial institutions high access fees. The Commission made it legally binding for S&P to abolish licensing fees paid by banks for using US ISINs.
- ⁴ For more details, see Angeliki Skoura,, Julian Presber, and Jang Schiltz's article, ["Luxembourg Fund Data Repository"](#), *Data* 5, no. 3 (July 2020): 62.
- ⁵ CUSIP and SEDOL are discussed in the next few sections. For details on WKN, refer to the [definition offered by Börse Frankfurt](#).
- ⁶ Example taken from the [ISIN organization web page](#).
- ⁷ Keep in mind that many financial identification systems are always expanding, so just because an identifier doesn't cover a certain market segment now doesn't imply it won't tomorrow.
- ⁸ The syndicated lending market is where a group of financial institutions (the syndicate) jointly extend a loan (often large loans) to a single borrower.
- ⁹ In finance, a swap is a type of derivative instrument in which two parties agree to exchange financial instruments, cash flows, or payments over a specified period of time.
- ¹⁰ To explore the utility of UTIs in financial markets, I recommend reading Swift's article on ["The Unique Transaction Identifier and Its Value in Securities Settlement"](#).
- ¹¹ Refer to BIS (editors): Committee on Payments and Market Infrastructures Board of the International Organization of Securities Commissions Technical Guidance, ["Harmonisation of the Unique Transaction Identifier"](#), February 2017.
- ¹² Sedol technical documentation is available on the [LSE website](#).
- ¹³ Information taken from the [official company page of HSBC on the LSE website](#).

14 An option is financial derivative instrument that gives its holder the right, but not the obligation, to buy or sell a specific quantity of an underlying asset at a given strike price on or before a specified future date, depending on the option style.

15 To explore for yourself, please use [this link](#).

16 More on blockchain internals will be discussed in [Chapter 8](#).

17 Here is an exercise: take the Luhn algorithm that we defined previously in the International Securities Identification Number and use it to validate a credit or debt card number (BE CAREFUL: don't use your own card number; a list of test credit card numbers is [available online](#)).